

Quiz 5-Solution

Sec D

Q1 Sum of chars in 1220 = $1+2+2 = 5$
1 / / / / 2111 = $2+1+1+1 = 5$
Hence, 1220 R 2111.

Q2 → Reflexivity: $\forall m \in \mathbb{Z}, (m-m) = 0$ which is divisible by 5.
Hence, $5 \mid (m-m)$ i.e. it is reflexive.

→ ^{Symmetry} Transitivity: Suppose mFn i.e. $5 \mid (m-n)$ i.e. $m-n=5k$; for some integer k

$$n-m = -5k$$

$$\text{or } n-m = 5(-k)$$

Here, $-k$ is an integer.

Thus, by definition $5 \mid (n-m)$.

Hence, if mFn then nFm . i.e. it is symmetric.

→ Transitivity: Suppose that mFn and nFp i.e.

$$5 \mid (m-n) \text{ and } 5 \mid (n-p)$$

$$\text{or } m-n=5k_1 \text{ and } n-p=5k_2; \text{ for some integers } k_1 \text{ and } k_2$$

$\hookrightarrow \textcircled{a} \qquad \qquad \qquad \hookrightarrow \textcircled{b}$

Adding eqs. $\textcircled{a}$ and $\textcircled{b}$

$$m - \cancel{n} = 5k_1$$

$$\cancel{n} - p = 5k_2$$

$$m-p = 5k_1 + 5k_2$$

$$m-p = 5(k_1+k_2)$$

Here, k_1+k_2 is an integer

$$\text{So, } 5 \mid (m-p)$$

Hence, if mFn and nFp , then mFp .

Date i.e. it is transitive.

Q3: (a) Yes, it is reflexive, as there are all 1's on the main diagonal of the matrix.

(b) It is ~~not~~ symmetric as $\text{matrix}(2,3)=1$ and $\text{matrix}(3,2)=1$ and ~~mat~~

(c) In the matrix, $\text{matrix}(2,3)=1$ and $\text{mat}(3,2)=1$ and so is $\text{mat}(2,2)=1$ and $\text{mat}(3,2)=1$ and $\text{mat}(2,3)=1$ and so is $\text{mat}(3,3)=1$.
Hence, it is transitive.